

BANARAS HINDU UNIVERSITY

B.Voc. Semester II Examination, 2023-24

Paper: BVCA-201 — Introduction To Databases

Time: Three Hours

Maximum Marks: 70

Instructions: Please attempt all Sections. Questions of Section A, B, and C carry 12, 6, and 1 mark each respectively.

SECTION — A

Note: Please answer in approximately 500 words each. Attempt any two questions from the following. Each question carries 12 marks.

Question 1. Describe the three-schema architecture. Why do we need mappings between schema levels? How do different schema definition languages support this architecture?

Question 2. Discuss the different types of user-friendly interfaces and the types of users who typically use each.

Question 3. Discuss the various reasons that lead to the occurrence of NULL values in relations.

Question 4. Discuss the role of a high-level data model in the database design process.

SECTION — B

Note: Please answer approximately in 200 words each. Each question carries 6 marks. Attempt any six from the following.

Question 5. Answer any six parts of the following:

- (a) What is the difference between logical data independence and physical data independence? Which one is harder to achieve? Why?
- (b) Discuss the different data types of SQL.
- (c) Discuss the entity integrity and referential integrity constraints. Why is each considered important?
- (d) Discuss Functional dependency with a suitable example.
- (e) What is a relationship type? Explain the differences among a relationship instance, a relationship type, and a relationship set.
- (f) What is a participation role? When is it necessary to use role names in the description of relationship types?
- (g) What is meant by a recursive relationship type? Give some examples of recursive relationship types.

- (h) Why are duplicate tuples not allowed in a relation?
- (i) What is the difference between a key and a superkey?

SECTION — C

Note: Please answer in the briefest possible way. Each question carries 1 mark.

Question 6. Define the following terms:

- (a) Data model
- (b) Database schema
- (c) Client/server architecture
- (d) Owner entity type
- (e) Weak entity type
- (f) Entity
- (g) Attribute Value
- (h) Relation state
- (i) Degree of a relation
- (j) Normalization